

1. LP.

$$[x, fval] = \text{linprog}(f, A, b, Aeq, beq, lb, ub, options)$$

$$[x, fval] = \text{intlinprog}(f, intcon, A, b, Aeq, beq, lb, ub, options)$$

LP标准型:

$$\begin{aligned} \min C^T x \\ \text{s.t. } Ax = b \\ x \geq 0 \end{aligned} \Leftrightarrow \begin{aligned} \min C^T x \\ \text{s.t. } Ax \leq b \\ x \in \mathbb{R}^n \end{aligned} \Leftrightarrow \begin{aligned} \max C^T x \\ \text{s.t. } Ax \geq b \\ x \in \mathbb{R}^n \end{aligned}$$

转换 ① $t \geq \max\{x_1, x_2\} \Leftrightarrow t \geq x_1, t \geq x_2$.

② $\|x\|_\infty = \max_{i=1, \dots, n} |x_i| \leq t \Rightarrow -t \leq x_i \leq t, i=1, \dots, n$

③ $\|x\|_1 = |x_1| + \dots + |x_n| \leq t \Rightarrow |x_i| \leq z_i, i=1, \dots, n, \sum_{i=1}^n z_i = t$.

可行集 $F_P := \{x \in \mathbb{R}^n, | Ax = b, x \geq 0\}$. $F_P \neq \emptyset \Leftrightarrow$ 原问题可行.

[Farkas引理] $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$, 则下列中必有一成立:

(i) $\exists x \geq 0$ s.t. $Ax = b$

(ii) $\exists y$ s.t. $y^T A \leq 0$ 且 $y^T b > 0$. (原问题不可行)

[证]. 反证. ①. $\forall x \geq 0$ s.t. $Ax = b$. 设 $\exists y$ s.t. $y^T A \leq 0$.

$\Rightarrow y^T b = y^T (Ax) = (y^T A)x \leq 0 \Rightarrow$ 不满足(ii).

② $\forall x \geq 0$ 有 $Ax < b \Rightarrow b \notin \{Ax | x \geq 0\}$ = 由A的列生成的锥K (K闭凸).

由分离超平面定理, $\exists y \in \mathbb{R}^m, \alpha \in \mathbb{R}$ s.t. $\forall z \in K, y^T z \leq \alpha$ 且 $y^T b > \alpha$

$0 \in K \Rightarrow y^T 0 = 0 \leq \alpha \Rightarrow \alpha \geq 0$.

K是锥 $\Rightarrow \forall z \in K, t \geq 0$ 有 $tz \in K \Rightarrow \forall z \in K, t \geq 0, y^T(tz) \leq \alpha$

$t \rightarrow \infty \Rightarrow y^T(tz) \rightarrow \infty$. 矛盾 $\Rightarrow \forall z \in K, y^T z \leq 0$.

$\exists y^T$ s.t. $y^T z \leq 0$.

$\Rightarrow$ 取 $\alpha = 0$ 时, 分离超平面定理有

$\forall z \in K, y^T z \leq 0$ 且 $y^T b > 0$.

$\forall z \in K, x \geq 0$ 有 $y^T Ax = y^T (Ax) = y^T z \leq 0$

可证 $y^T A$ 每行 $\leq 0 \Rightarrow y^T A \leq 0$.

最优值 $p^* = \begin{cases} < \infty \rightarrow \text{有解} \\ = +\infty \rightarrow \text{不可行} \\ = -\infty \rightarrow \text{无界问题} \end{cases}$

$\Rightarrow$ Lagrangian function $L: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}_+^m \rightarrow \mathbb{R}$. $\left\{ \begin{array}{l} \text{Lagrangian multiplier } y \in \mathbb{R}^m \\ \text{dual variable } s \in \mathbb{R}_+^m \end{array} \right.$

$L(x, y, s) = C^T x + y^T (b - Ax) - s^T x$.

$Ax \leq b \Rightarrow +y^T (Ax - b)$

($y \geq 0$)

$\Rightarrow$ dual function. $g: \mathbb{R}^m \times \mathbb{R}_+^m \rightarrow \mathbb{R}$.

$g(y, s) = \min_x L(x, y, s) = \min_x x^T (c - A^T y - s) + b^T y$.

$= \begin{cases} b^T y & c - A^T y - s = 0 \\ -\infty & c - A^T y - s \neq 0 \end{cases}$

$\Rightarrow D, \max_{y \in \mathbb{R}^m} b^T y = \max_y g(y, s) = \max_y \min_x L(x, y, s)$

s.t. $c - A^T y = s$

$s \geq 0$.

$\Leftrightarrow \max_y b^T y$
s.t. $c - A^T y \geq 0$

最优值 $d^* = \begin{cases} < \infty \rightarrow \text{有解} \\ = -\infty \rightarrow \text{不可行} \\ = +\infty \rightarrow \text{无界} \end{cases}$

[Weak duality] $d^* \leq p^*$.

[P]. $\forall x \in F_P, (y, s) \in d$.

$$b^T y = (Ax)^T y = y^T (A^T y) = x^T (c - s) = c^T x - s^T x \leq c^T x.$$

$$\Rightarrow d^* = \max_{y \in d} b^T y = \min_{x \in F_P} c^T x = p^*$$

推论: $b^T y = c^T x$ 时 $x, (y, s)$ 是强对偶对.

[Strong duality] d^*, p^* 至少一个有限 $\Rightarrow d^* = p^*$.

[P]. 设 $p^* < \infty, \forall \varepsilon > 0, \begin{cases} Ax = b \\ c^T x = (p^* - \varepsilon) \end{cases}$ 无解 $\Leftrightarrow \begin{pmatrix} A \\ -c^T \end{pmatrix} x = \begin{pmatrix} b \\ -(p^* - \varepsilon) \end{pmatrix}$ 无解.

由 Farkas 引理, $\exists z \in \mathbb{R}^{n+1}$ s.t. $(A, -c^T)z \leq 0$ 且 $(b^T - (p^* - \varepsilon))z > 0$.

设 $z = \begin{pmatrix} y \\ y_0 \end{pmatrix}, y_0 \in \mathbb{R}, y \in \mathbb{R}^m, \Rightarrow$ 必有 $y_0 > 0$.

$$\text{w.l.o.g. 设 } y_0 = 1, \Rightarrow \begin{cases} Ay - c \leq 0 \\ By - (p^* - \varepsilon) > 0 \end{cases} \Rightarrow \begin{cases} c - Ay \geq 0 \\ b^T y - (p^* - \varepsilon) > 0 \end{cases} \Rightarrow y \in F_d$$

$$\varepsilon \rightarrow 0, b^T y - p^* > 0 \Rightarrow b^T y \geq p^* \Rightarrow d^* \geq p^*$$

$$\text{由弱对偶有 } d^* \leq p^* \Rightarrow d^* = p^*.$$

[Primal & dual Farkas 引理].

①. 原问题不可行 $\Leftrightarrow \exists y$ s.t. $A^T y \leq 0$ 且 $b^T y > 0$.

②. 对偶问题不可行 $\Leftrightarrow \exists x \geq 0$ s.t. $Ax = 0$ 且 $c^T x < 0$.

基本可行点: $\Rightarrow$ 满足 $x \in F_P, \forall i \in \{1, \dots, n\} \setminus B, x_i = 0, A_B$ 非奇异.

[LP 基本定理 1].

①. LP 有可行点 $\Rightarrow$ 有基本可行点.

②. LP 有解 $\Rightarrow$ 至少 1 解为基本可行点最优.

顶点: $\Rightarrow x \in P$ 不能被 P 中其它两点线性表示. i.e. $\nexists y, z \in P$ s.t. $x = sy + (1-s)z, s \in (0, 1)$.

[LP 基本定理 2]. 顶点 $\Leftrightarrow$ 基本可行点.

P 到 P 映射. dual. $\max_y b^T y$
s.t. $c - A^T y = s$
 $s \geq 0, y \in \mathbb{R}^m$. 求 $x \in \mathbb{R}^n$

$$\Rightarrow L: \mathbb{R}^m \times \mathbb{R}_+^n \rightarrow \mathbb{R}$$

$$L(y, x) = b^T y + x^T (c - A^T y)$$

$$\Rightarrow f(x) = \max_y L(y, x)$$

$$= \max_y y^T (b - Ax) + c^T x$$

$$= \begin{cases} c^T x & Ax = b \\ \infty & Ax \neq b \end{cases}$$

$$\Rightarrow d, \text{ dual. } \min_x f(x) = \min_x \max_y L(y, x) = \min_x c^T x$$

$$\text{s.t. } Ax = b, x \geq 0.$$

2.

zero-sum := sum of net winnings = 0.

dominated := better strategy regardless opponent's choice.

value of the game := the payoff to player 1 when both players play optimally.

fair := when value of the game = 0.

pay off matrix.
(零和).

	1	2	3	Min
1	5	0	6	0
2	3	4	2	2
3	6	2	1	1
Max	5	4	6	

player 1 → 极大极小.

player 2 → 极小极大.

Solution stable := 在对方固定该策略下, 自己更改策略不能再收益更高.

Saddle point := maximin = minimax value

player 1:

lower value $A^- := \max_i \min_j a_{ij}$

upper value $A^+ := \min_j \max_i a_{ij}$.

$A^- = A^+ = \text{value of the game.}$

stable. $\Leftrightarrow a_{pq} = \min_j a_{pj} = \max_i a_{iq}$
saddle point.

Theo 1. $A^- \leq A^+$.

[i] $\forall k \in \{1, \dots, n\}, \min_j a_{ij} \leq a_{ik} \Rightarrow A^- = \max_i \min_j a_{ij} \leq \max_i a_{ik}$.

$\Rightarrow A^- = \max_i \min_j a_{ij} = \min_k (\max_i \min_j a_{ij}) \leq \min_k (\max_i a_{ik}) = A^+$.

Theo 2. $A \in \mathbb{R}^{m \times n}$ stable $\Leftrightarrow$ has saddle point.

[i] $\Rightarrow$. A stable, i.e. $a_{pq} = A^- = A^+$.

$a_{pq} = A^- = \max_i \min_j a_{ij} \Rightarrow p \in \arg \max_i (\min_j a_{ij}) \Rightarrow a_{pq} = \min_j a_{pj}$

$a_{pq} = A^+ = \min_j \max_i a_{ij} \Rightarrow q \in \arg \max_j (\min_i a_{ij}) \Rightarrow a_{pq} = \max_i a_{iq}$

$\Leftrightarrow$. i.e. $a_{pq} = \max_i a_{iq} = \min_j a_{pj}$ is saddle point. $\Rightarrow a_{pq}$ saddle point.

$A^+ = \min_j (\max_i a_{ij}) \leq \max_i a_{iq} = a_{pq} = \min_j a_{pj} \leq \max_i (\min_j a_{ij}) = A^-$

$\Rightarrow A^+ = A^-$, stable.

Theo 3. dominance 消除不影响鞍点.

混合策略.

stable 时, player 1 策略 $X = \{x \in \mathbb{R}^m : x \geq 0, \sum_{i=1}^m x_i = 1\}$.

player 2 策略 $Y = \{y \in \mathbb{R}^n : y \geq 0, \sum_{j=1}^n y_j = 1\}$.

expected payoff (player 1): $X^T A y$.

(player 2): $-X^T A y$

player 1 极大极小
player 2 极小极大

$\Rightarrow$ 最优策略 X^* max v
s.t. $A^T X \geq (v, \dots, v)^T$
(1, ..., 1) $X = 1$
 $X \geq 0, v \in \mathbb{R}$.

最优策略 Y^* min w
s.t. $A y \leq (w, \dots, w)^T$
(1, ..., 1) $y = 1$
 $y \geq 0, w \in \mathbb{R}$.

Theo 4. 都用 optimal 混合策略时, ^{optimal} expected payoff $p^* = \max_x \min_y x^T A y$
 $=$ 对于 optimal expected loss $d^* = \min_y \max_x x^T A y$
此时 stable.
 $d^* = \min_x \max_y y^T A x$

图 (V, E) . V : 节点 node.

E : 边 edge

connected: 共享同一个端点的两边.

path: 边连起来的一条通路.

network: 图 (V, E) 且有权重 $w: E \rightarrow \mathbb{R}$.

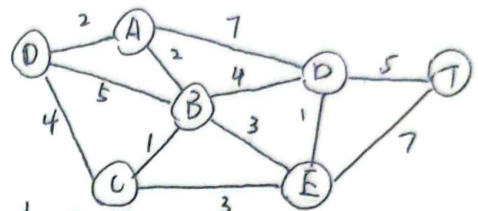
network cut. $N = (V, E)$ 的节点子集 $K \subseteq V$, 补集 $\bar{K} = V \setminus K$ 满足 $\begin{cases} K \cap \bar{K} = \emptyset \\ K \cup \bar{K} = V \end{cases} \Rightarrow \text{cut} := (K, \bar{K})$
 cut-set 割集: $C(N, K) = \{(a, b) \in E: a \in K, b \in \bar{K}\}$.

Dijkstra's algorithm.

$d = (0, \infty, \dots, \infty)$. distance.

$P = (\emptyset, \dots, \emptyset)$. previous-node

$V = (0, \dots, 0)$. visited-node vectors.



	d	P	V
1	0		0
2	∞	0	0
3	∞	0	0
4	∞	0	0
5	∞	0	0
6	∞	0	0
7	∞	0	0

(初始)

	d	P	V
1	0		1
2	2	1	0
3	5	1	0
4	4	1	0
5	∞	0	0
6	∞	0	0
7	∞	0	0

(在 O)

	d	P	V
1	0		1
2	2	1	1
3	4	2	0
4	4	1	0
5	8	3	0
6	7	3	0
7	∞	0	0

基于目前的 d 的路径计算出来
 0 到 A 是 2, 0 到 C 是 4, 所以仍为 4.
 从 0 到 A (到 A)
 到 D = 9

	d	P	V
1	0		1
2	2	1	1
3	4	2	1
4	4	1	1
5	8	3	0
6	7	3	1
7	14	6	0

(到 E)

用上步所得的目前 $d_E = 7$ 去更新才得到 14.

	d	P	V
1	0		1
2	2	1	1
3	4	2	1
4	4	1	0
5	8	3	0
6	7	3	0
7	∞	0	0

(到 B)

	d	P	V
1	0		1
2	2	1	1
3	4	2	1
4	4	1	1
5	8	3	0
6	7	3	0
7	∞	0	0

(到 C)

	d	P	V
1	0		1
2	2	1	1
3	4	2	1
4	4	1	1
5	8	3	1
6	7	3	1
7	14	6	0

(到 D)

	d	P	V
1	0		1
2	2	1	1
3	4	2	1
4	4	1	1
5	8	3	1
6	7	3	1
7	13	5	1

✓ (到 T)

到 T 的 d 是 8, 再加上 D 到 T 为 5.

$\Rightarrow$ 倒推: $T \rightarrow D \rightarrow B \rightarrow A \rightarrow O$
 i.e. 最短路径为 $O \rightarrow A \rightarrow B \rightarrow D \rightarrow T$.

3.

无约束优化. $f: \mathbb{R}^n \rightarrow \mathbb{R}$. $\min_x f(x)$.

local minimizer $:= \forall x \in B(x^*, \varepsilon) \in \mathbb{R}^n, f(x^*) \leq f(x)$

global minimizer $:= \forall x \in \mathbb{R}^n, f(x^*) \leq f(x)$

stationary point $:= \nabla f(x) = 0$.

[1阶必要]. f 连续可微, x^* 局部极小 $\Rightarrow \nabla f(x^*) = 0$.

证. 反设 $-\nabla f(x^*) \neq 0 \Rightarrow -\nabla^T f(x^*) \nabla f(x^*) = -\|\nabla f(x^*)\|^2 < 0$.

f 连续可微 $\Rightarrow \forall x \in B(x^*, \rho)$.

$f(x) = f(x^* + tp) = f(x^*) + \nabla^T f(x^* + tp) p < f(x^*)$ 矛盾.

[2阶必要]. f 2次连续可微, x^* 局部极小 $\Rightarrow \nabla f(x^*) = 0, \nabla^2 f(x^*) \geq 0$.

[2阶充分]. f 2次连续可微, $\nabla f(x^*) = 0, \nabla^2 f(x^*) > 0 \Rightarrow x^*$ 严格局部极小.

凸方程 $:= f: \mathbb{R}^n \rightarrow \mathbb{R}$ 满足 $\forall x_1, x_2 \in \mathbb{R}^n, t \in [0, 1]$,
 $f(x_1 t + (1-t)x_2) \leq t f(x_1) + (1-t) f(x_2)$

[凸理论]. f 凸

$\Rightarrow$ ①. 局部极小 $\Leftrightarrow$ 全局极小.

②. f 可微时 $\nabla f(x^*) = 0 \Rightarrow x^*$ 全局极小 / 局部极小.

线搜索. $x_{k+1} = x_k + \overset{\text{步长 step}}{s_k} p_k$.

$s_k = \arg \min_{s \geq 0} f(x_k + s p_k)$.

1. $s \nabla$. $p_k = -\nabla f(x_k)$.

2. Newton. $(\nabla^2 f(x_k) > 0)$. $p_k = -[\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$.

Wolfe 条件. [Armijo 条件]. $c_1 \in (0, 1)$, s_k 满足 $f(x_k + s_k p_k) \leq f(x_k) + s_k (c_1 \nabla^T f(x_k) p_k)$.

[Curvature 条件]. $c_2 \in (c_1, 1)$. s_k 满足 $\nabla^T f(x_k + s_k p_k) \geq c_2 \cdot \nabla^T f(x_k) p_k$.

Taylor.

f 连续可微, $f(x+p) = f(x) + \nabla f(x+p)^T p, t \in (0, 1)$

f 2次连续可微. $f(x+p) = f(x) + \nabla f(x)^T p + \frac{1}{2} p^T \nabla^2 f(x+tp) p, t \in (0, 1)$.

4. 约束优化. $f: \mathbb{R}^n \rightarrow \mathbb{R}$. $c_i: \mathbb{R}^n \rightarrow \mathbb{R}$. s.t. $\min_x f(x)$
 $c_i(x) = 0, i \in E$ (equality)
 $c_i(x) \geq 0, i \in I$ (inequality).

可行集 $F = \{x \in \mathbb{R}^n \mid c_i(x) = 0, \forall i \in E; c_i(x) \geq 0, \forall i \in I\}$.

积极集 Active set := 可行点 x 的积极集为 $A(x) = E \cup \{i \in I, c_i(x) = 0\}$.

LLCQ 在 $x \in F$ 成立 := $\nabla c_i(x), i \in A(x)$ 互相 L.I.

KKT, x^* 局部最优且 LLCQ, 则有

KKT 条件. $x^* \in F, \lambda^* = \begin{pmatrix} \lambda^* \\ y^* \end{pmatrix} = \nabla f(x^*) - \sum_{i=1}^m y_i^* \nabla c_i(x^*) = 0$ ① Lagrangian 条件
 $c_i(x^*) = 0, \forall i \in E, c_i(x^*) \geq 0, \forall i \in I$. ② 原问题的 KKT 条件
 $y_i^* \geq 0, \forall i \in I$. ③ 对偶问题的 KKT 条件
 $y_i^* c_i(x^*) = 0, \forall i \in E \cup I$. ④ 互补条件

二次罚函数.

$$Q(x, p) = f(x) + \frac{p}{2} \sum_{i \in E} c_i^2(x) + \frac{p}{2} \sum_{i \in I} (\min(c_i(x), 0))^2, \quad p > 0 \text{ 罚参数}$$

增广 Lagrangian. $\rightarrow$ 解 ∇ 等式约束.

$$L_A(x, y, p) = f(x) + \frac{p}{2} \sum_{i \in E} c_i^2(x) - \sum_{i \in E} y_i c_i(x)$$

Backtracking. $v > 0, r \in (0, 1), c \in (0, 1)$.

设 $s_k = v$. $f(x_k + s_k p_k) > f(x_k) + s_k \cdot (c_1 \nabla f(x_k)) \cdot p_k$ 时有 $s_k = r \cdot s_k$.

[Zoutendijk]. $\{x_k\}$ 由基于下降方向 $\{p_k\}_{k \geq 0}$ 和步长 $\{s_k\}_{k \geq 0}$ 生成的线性点, 且满足

Wolfe condition. $f: \mathbb{R}^n \rightarrow \mathbb{R}$ Lipschitz 连续可微. $\{a_k\}_{k \geq 0}$ 与 p_k 和 $\nabla f(x_k)$ 之间夹角,

$$\text{则有 } \sum_{k=0}^{\infty} \cos^2(\alpha_k) \|\nabla f(x_k)\| < \infty.$$

5. 排队.

[Little Law]. 排队系统 $[0, T]$, $T \in (0, \infty)$

①. $t=0, t=T$ 为宜, 则 $L = \bar{\lambda} W$.

②. $L = \bar{\lambda} W$ (初末时刻系统人数不为宜时).

$N(t) := t$ 时排队系统中人数.

$\bar{\lambda} :=$ 平均到达率.

$L :=$ 系统中平均人数.

$W :=$ 人平均等待时间.

Exp. Dis. $\alpha > 0$. $f_T(t) = \begin{cases} \alpha \cdot e^{-\alpha t} & t \geq 0 \\ 0 & t < 0 \end{cases}$

性质. ①. $\forall t \geq 0, f_T(t) \downarrow$. $\forall dt > 0, P[T \in [a, a+dt]] > P[T \in [t, t+dt]]$.

②. 无记忆性. $\forall t, dt > 0, P[T > t+dt | T > t] = P[T > dt]$.

③. $T_i \sim \text{Exp}(\alpha_i), i=1, \dots, n$, i.i.d.

$\Rightarrow U = \min(T_1, \dots, T_n) \sim \text{Exp}(\sum_{i=1}^n \alpha_i)$.

④. $T \sim \text{Exp}(\alpha) :=$ 每次事件发生后到下一次等待时间.

$\Rightarrow X(t) \sim \text{Poi}(\alpha t) :=$ 到时间 t 为止事件发生总次数. $P[X(t)=n] = \frac{(\alpha t)^n \cdot e^{-\alpha t}}{n!}$

MM1 $\rightarrow$ 研究 $N(t) := t$ 时系统中人数.

$\rightarrow$ 到达间隔 $I \sim \text{Exp}(\lambda) \Rightarrow$ 到 t 时为止到达次数 $N_I(t) \sim \text{Poi}(\lambda t)$.

$\rightarrow$ 服务时间 $S \sim \text{Exp}(\mu) \Rightarrow$ 到 t 时为止服务次数 $N_S(t) \sim \text{Poi}(\mu t)$.

服务台 μ . 每时间服务 μ 人. 平均服务 1 人花费 $\frac{1}{\mu}$ 时间.

只有 1 个服务台.

[Kolmogorov 方程]. $P_n(t) = P[N(t)=n]$. s.t. $P_i(t) \geq 0, P_0(t) + \dots = 1$.

设队上限 $M-1$.

$$\begin{cases} \frac{d}{dt} P_0(t) = P_1(t) \cdot \mu - P_0(t) \cdot \lambda \\ \frac{d}{dt} P_n(t) = P_{n-1}(t) \cdot \lambda - P_n(t) \cdot (\lambda + \mu) + P_{n+1}(t) \cdot \mu, \quad 1 \leq n \leq M. \end{cases}$$

$$L(t) = E[N(t)] = \sum_{n=0}^M n \cdot P_n(t).$$

稳态. i.e. $\frac{d}{dt} P_n = 0, P_n \rightarrow P_n(t), \Rightarrow \begin{cases} P_0 \cdot \lambda = P_1 \cdot \mu \\ P_n(\lambda + \mu) = P_{n-1} \cdot \lambda + P_{n+1} \cdot \mu \end{cases}$

$M \rightarrow \infty$ 时队无上限, $P_n = \left(\frac{\lambda}{\mu}\right)^n \cdot P_0$. $\rho = \frac{\lambda}{\mu}$ service facility utilization factor. $\rho < 1$.

$M/M/1$ 有稳态分布 $\Leftrightarrow \rho < 1$.

Exp. $\alpha > 0, f_T(x) = \begin{cases} \alpha \cdot e^{-\alpha x} & x \geq 0 \\ 0 & x < 0 \end{cases}$

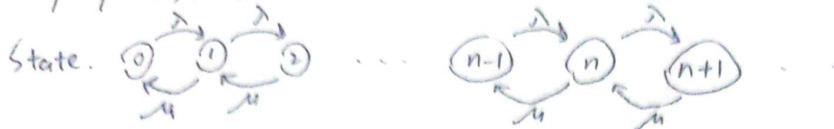
$$\forall a, b \in \mathbb{R}, P[T \in [a, b]] = \int_a^b f_T(x) dx.$$

$$X(t) \sim \text{Poi}(\alpha t) \Rightarrow E[X(t)] = \alpha t.$$

FIFO 的 $M/M/1$ 中顾客等待时间 $w \sim \text{Exp}(\mu - \lambda)$, $\mu > \lambda$, 平均等待时间

$$W = E[w] = \frac{1}{\mu - \lambda}.$$

$M/M/1$ 生灭过程.



一般生灭过程.

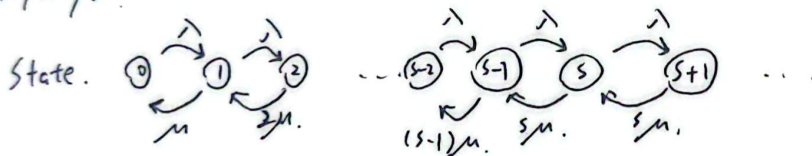


$$\begin{cases} \frac{d}{dt} p_0(t) = p_1(t) \cdot \mu_1 - p_0(t) \cdot \lambda_0 \\ \frac{d}{dt} p_n(t) = p_{n-1}(t) \cdot \lambda_{n-1} - p_n(t) \cdot (\lambda_n + \mu_n) + p_{n+1}(t) \cdot \mu_{n+1}, \quad n \geq 1. \end{cases}$$

稳态. $\begin{cases} p_1 \mu_1 = p_0 \lambda_0. \\ p_{n-1} \lambda_{n-1} + p_{n+1} \mu_{n+1} = p_n \cdot (\lambda_n + \mu_n), \quad n \geq 1 \end{cases}$

$$n > 0, C_n = \frac{\lambda_{n-1} \lambda_{n-2} \dots \lambda_0}{\mu_n \dots \mu_1}, \quad p_0 = \left(1 + \sum_{n=1}^{\infty} C_n \right)^{-1}, \quad p_n = C_n \cdot p_0.$$

$M/M/s \rightarrow \mu_n = \min(n, s) \cdot \mu.$



$$C_n = \begin{cases} \frac{1}{n!} \cdot \left(\frac{\lambda}{\mu} \right)^n, & n \leq s \\ \frac{1}{s! s^{n-s}} \cdot \left(\frac{\lambda}{\mu} \right)^n, & n > s. \end{cases}$$

$$p_0 = \left(\sum_{n=0}^{s-1} \frac{1}{n!} \cdot \left(\frac{\lambda}{\mu} \right)^n + \frac{\left(\frac{\lambda}{\mu} \right)^s}{s! \left(1 - \frac{\lambda}{s\mu} \right)} \right)^{-1}$$

$$L_q = \sum_{n=s}^{\infty} (n-s) \cdot p_n = \frac{p_0}{s!} \cdot \left(\frac{\lambda}{\mu} \right)^s \cdot \frac{\left(\frac{\lambda}{s\mu} \right)}{\left(1 - \frac{\lambda}{s\mu} \right)^2}.$$

$$W_q = \frac{L_q}{\lambda}.$$

$$W = W_q + W_s = W_q + \frac{1}{\mu}.$$

$$L = \lambda W = \lambda \left(W_q + \frac{1}{\mu} \right) = L_q + \frac{\lambda}{\mu}.$$

$$M/M/1/K \rightarrow \text{MMI 但上限 } N(t) = K \Rightarrow \begin{cases} \lambda_n = \begin{cases} \lambda, & n = 0, \dots, K-1 \\ 0, & n = K \end{cases} \\ \mu_n = \begin{cases} \mu, & n = 1, \dots, K \\ 0, & n = 0 \end{cases} \end{cases}$$

$$\rho = \frac{\lambda}{\mu}, \quad p_0 = \begin{cases} \frac{1-\rho}{1-\rho^{K+1}}, & \rho \neq 1 \\ \frac{1}{K+1}, & \rho = 1 \end{cases}$$

$$p_n = \left(\frac{\lambda}{\mu}\right)^n \cdot p_0 = \rho^n \cdot p_0, \quad n = 0, \dots, K.$$

$$L = \begin{cases} \frac{\rho}{1-\rho} - \frac{(K+1) \cdot \rho^{K+1}}{1-\rho^{K+1}}, & \rho \neq 1 \\ \frac{K}{2}, & \rho = 1 \end{cases}$$

$$W = \frac{L}{\lambda} = \begin{cases} \frac{1}{\lambda \cdot (1-\rho)} \cdot \left(\frac{\rho}{1-\rho} - \frac{(K+1) \cdot \rho^{K+1}}{1-\rho^{K+1}} \right), & \rho \neq 1 \\ \frac{K}{2\lambda \cdot (1-\rho_0)}, & \rho = 1. \end{cases}$$